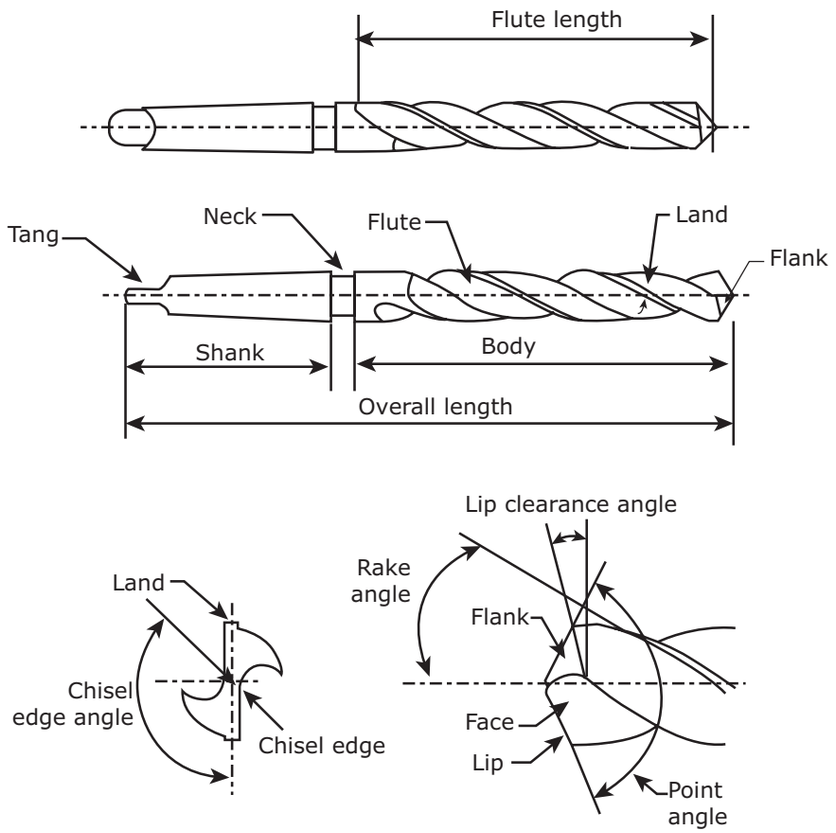




Twist Drill Nomenclature

Flutes

The grooves in the body of the drill are known as flutes.

- Flutes form the cutting edges on the point.
- To allow the chips to escape
- To cause the chips to curl
- To permit the cutting fluid to reach the cutting edges.

Angles

Chisel edge Angle

The obtuse angle included between the chisel edge and the lips as viewed from the end of the drill. It's usual value of this angle various from 120^0 to 135^0 Helix angle (or) rake angle.

Point angle

This is the angle included between the two lips projected upon a plane parallel to the drill axis and parallel to the two cutting lips. The usual point angle is 118^0 when hard alloys are drilled the value of angle increases.

Lip clearance Angle

The angle formed by the flank and a plane at right angle to the drill axis. The angle is normally measured at the periphery of the drill. The lip clearance angle ranges from 12^0 to 15^0.

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